

Mathematical Foundations of Political Methodology

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Random Variables

- CDF
- Continuous Random Variables
- More on Expectation and Variance
- Brief introduction to inference

Cumulative Distribution Function

Definition

The Cumulative Distribution Function (CDF) of a random variable X is:

$$F_X(x) = P(X \leq x), \quad -\infty < x < \infty$$

$$F_X(x) = \sum_{k \leq x} p_X(k)$$

The CDF characterizes how probability cumulates as X gets larger.

Cumulative Distribution Function: Example

Suppose $X \sim \text{Binomial}(n = 10, p = .3)$ What is the probability that $x \leq 4$?

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What is $F_X(22)$?

What is $F_X(-1)$?

Cumulative Distribution Function

Theorem

$F(x)$ is a CDF if and only if:

1 $F_X(x)$ is nondecreasing as x increases:

$$x_1 < x_2 \Rightarrow F(x_1) \leq F(x_2)$$

2 $F_X(x)$ has limits at $\pm\infty$

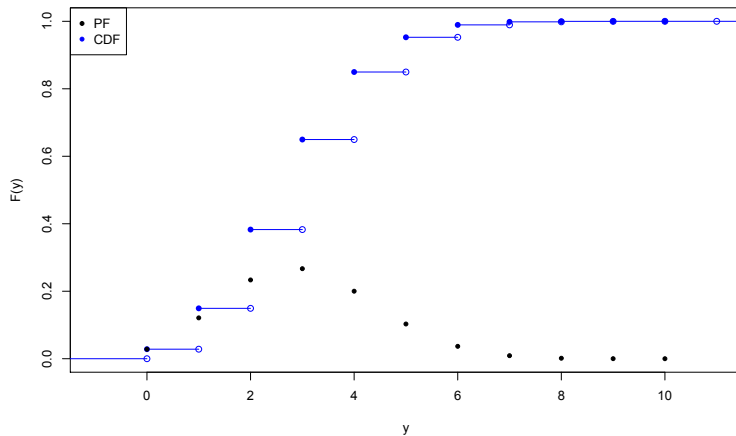
$$\lim_{x \rightarrow -\infty} F_X(x) = 0$$

$$\lim_{x \rightarrow \infty} F_X(x) = 1$$

3 Continuity from the right:

$$F_X(x) = \lim_{y \rightarrow x, y > x} F(y) \quad \forall x$$

CDF



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Expected value of a function of a random variable: Suppose X is a discrete random variable that takes on values x_i , $i = \{1, 2, \dots\}$, with probabilities $p(x_i)$.

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$$E[g(X)] = \sum_i g(x_i)p(x_i)$$

Functions of Random Variables: Corollary

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*Suppose X is a random variable and a and b are **constants** (not random variables). Then,*

$$E[aX + b] = aE[X] + b$$

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Proof.

$$E[aX + b] = \sum_{x:p(x)>0} (ax + b)p(x)$$

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Proof.

$$\begin{aligned} E[aX + b] &= \sum_{x:p(x)>0} (ax + b)p(x) \\ &= \sum_{x:p(x)>0} axp(x) + \sum_{x:p(x)>0} bp(x) \end{aligned}$$

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Famous Distributions

- Bernoulli (coin flips)
- Binomial (sums of coin flips)
- Multinomial (multivariate coin flips)
- Poisson (sums of events over time or space)

Models of how world works.

Poisson Distribution

Definition

Suppose X is a random variable that takes on values $X \in \{0, 1, 2, \dots\}$ and that $P(X = k) = p(k)$ is,

$$p(k) = e^{-\lambda} \frac{\lambda^k}{k!}$$

for $k \in \{0, 1, \dots\}$ and 0 otherwise. Then we will say that X follows a *Poisson* distribution with *rate* parameter λ .

$$X \sim \text{Poisson}(\lambda)$$

Poisson Distribution

Properties:

1) It is a probability distribution.

■ Recall the **Taylor expansion** of e^x

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

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■ Now expand the pdf of the poisson distribution:

$$e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{-\lambda} \left(1 + \lambda + \frac{\lambda^2}{2!} + \dots \right)$$

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$$\begin{aligned} e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} &= e^{-\lambda} \left(1 + \lambda + \frac{\lambda^2}{2!} + \dots \right) \\ &= e^{-\lambda} (e^{\lambda}) = 1 \end{aligned}$$

Poisson Distribution

- 2) $E[X] = \lambda$, that is, the parameter of the poisson distribution is its expectation

$$\begin{aligned} E[X] &= e^{-\lambda} \sum_{k=0}^{\infty} k \frac{\lambda^k}{k!} \\ &= e^{-\lambda} \lambda \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} \end{aligned}$$

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$$\begin{aligned} E[X^2] &= \lambda e^{-\lambda}(\lambda e^{\lambda} + e^{\lambda}) \\ &= \lambda(\lambda + 1) \end{aligned}$$

$$\text{var}(X) = E[X^2] - E[X]^2 = \lambda^2 + \lambda - \lambda^2 = \lambda$$

Joint PMFs

If we have more than one random variable, we can study their joint behavior.

If (x, y) are possible values of random variables X and Y , then

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Joint PMFs

If we want to know the probability of a set of joint events $(x, y) \in A$

$$P((X, Y) \in A) = \sum_{(x, y) \in A} p_{X,Y}(x, y)$$

We can also calculate the pmfs of X and Y individually (these are the marginal distributions:

$$p_X(x) = \sum_y p_{X,Y}(x, y), \quad p_Y(y) = \sum_x p_{X,Y}(x, y)$$

Example of Joint PMF

$$\Omega_1 = \{\text{Non-Democracy}, \text{ Democracy}\}$$

$$X(\omega) = \begin{cases} 0 & \text{if } \omega = \text{Non-Democracy} \\ 1 & \text{if } \omega = \text{Democracy} \end{cases}$$

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$p_{X,Y}$	0	1	p_Y
1	0.034	0.005	0.039
0	0.466	0.495	0.961
p_X	0.5	0.5	1

Linearity

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Conditional Probability Distribution Function

Definition

Suppose X and Y are discrete random variables with joint pdf $p(x, y)$. Then define the *conditional probability function* $p(x|y)$ as

$$p(x|y) = \frac{p(x, y)}{p_Y(y)}$$

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$$p(x_j) = \sum_{i=1}^N p(x_j|y_i)p(y_i)$$

A Table

	$Y = 0$	$Y = 1$	
$X = 0$	$p(0,0)$	$p(0, 1)$	$p_X(0)$
$X = 1$	$p(1,0)$	$p(1,1)$	$p_X(1)$
	$p_Y(0)$	$p_Y(1)$	

$$p_X(0) = p(0|y=0)p(y=0) + p(0|y=1)p(y=1)$$

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	$Y = 0$	$Y = 1$	
$X = 0$	0.01	0.05	?
$X = 1$	0.25	0.69	?
	0.26	0.74	

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$$\begin{aligned}p_X(0) &= p(0|y=0)p(y=0) + p(0|y=1)p(y=1) \\&= \frac{0.01}{0.26} \times 0.26 + \frac{0.05}{0.74} \times 0.74 \\&= 0.06\end{aligned}$$

$$\begin{aligned}p_X(1) &= p(1|y=0)p(y=0) + p(1|y=1)p(y=1) \\&= \frac{0.25}{0.26} \times 0.26 + \frac{0.69}{0.74} \times 0.74\end{aligned}$$

A Table

	Y = 0	Y = 1	
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*If X and Y are not independent, we will say they are **dependent***

Conditional Distribution

If X and Y are independent, then

$$\begin{aligned} p_{X|Y}(x|y) &= \frac{p(x, y)}{p_Y(y)} \\ &= \frac{p_X(x)p_Y(y)}{p_Y(y)} \\ &= p_X(x) \end{aligned}$$

In words: the distribution of X does not change as levels of Y change.

Conditional Expectation

$$\mathbb{E}(Y|X) = \sum_y y \cdot p_{Y|X}(y|x)$$

$\mathbb{E}(Y|X)$ is the *best* way to predict Y given X .

Continuous Random Variables

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We will need continuous random variables to assign numbers to these events.

Probability of a continuous random variable

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$$\int_a^b f_X(x) dx = P(X \in (a, b))$$

Cumulative Distribution Function

Cumulative distribution functions fully characterize continuous random variables.

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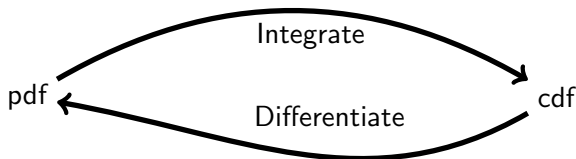
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RV: continuous uniform

Definition

We can define Y to have a uniform distribution on the interval (a, b)

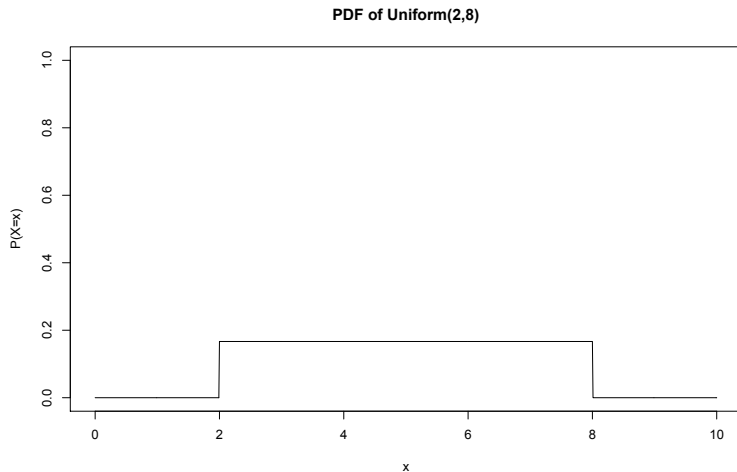
$$f_Y(y) = \begin{cases} \frac{1}{b-a} & \text{if } a < y < b \\ 0 & \text{otherwise} \end{cases}$$

Example: Uniform

Example: Suppose that we are waiting for Comcast to show up and install our cable package. They say that they may arrive between 2:00 and 8:00. Without any further information, you may have no reason to suspect any particular time over any other.

$$X \sim U(2, 8)$$

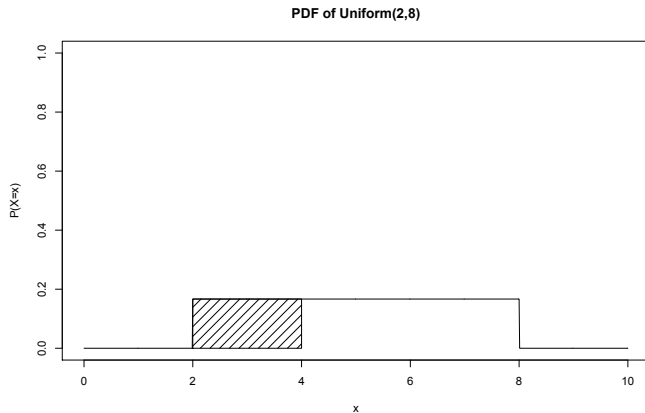
RV: continuous uniform



RV: continuous uniform

What is the probability that the cable installation truck arrives before 4:00?

RV: continuous uniform ($P(X \leq 4)$)



RV: continuous uniform

$$P(Y \leq y) = \int_{x=-\infty}^y f(x) dx = \int_{x=-\infty}^y \frac{1}{b-a} dx = \int_{x=a}^y \frac{1}{b-a} dx$$

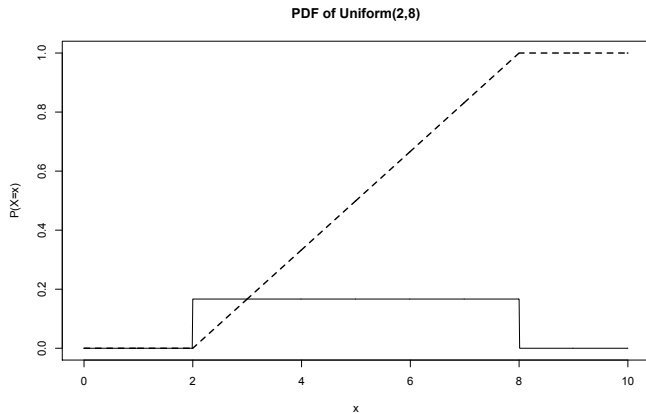
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Common Continuous Distributions: Exponential

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$$f(x) = \lambda e^{-\lambda x}$$

with $f(x) = 0$ for $x < 0$

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with $f(x) = 0$ for $x < 0$

λ is known as the 'rate' parameter.

$$\lambda = \frac{1}{E(X)}$$

Exponential CDF

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$$\begin{aligned} &= -\frac{\lambda e^{-\lambda x}}{\lambda} - \left(-\frac{\lambda e^{-\lambda \cdot 0}}{\lambda}\right) \\ &= 1 - e^{-\lambda x} \end{aligned}$$

Expectation With Continuous Random Variables

Definition

If X is a continuous random variable then,

$$E[X] = \int_{-\infty}^{\infty} xf(x)dx$$

Suppose $X \sim \text{Uniform}(0, 1)$. What is $E[X]$?

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Expectations of Functions

Proposition

Suppose X is a continuous random variable and $g : \mathbb{R} \rightarrow \mathbb{R}$. Then,

$$E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x)dx$$

Expectations of Functions

Suppose $g(X) = X^2$ and $X \sim \text{Uniform}(0, 1)$. What is $E[g(X)]$?

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Suppose X is a continuous random variable. Then,

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Proof.

$$E[aX + b] = \int_{-\infty}^{\infty} (ax + b)f(x)dx$$

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Definition

Variance. If X is a continuous random variable, define its variance, $\text{Var}(X)$,

$$\begin{aligned}\text{Var}(X) &= E[(X - E[X])^2] \\ &= \int_{-\infty}^{\infty} (x - E[X])^2 f(x) dx \\ &= E[X^2] - E[X]^2\end{aligned}$$

Variance: Random Variable

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$$\begin{aligned} \text{Var}(X) &= E[X^2] - E[X]^2 \\ &= \frac{1}{3} - \frac{1}{4} = \frac{1}{12} \end{aligned}$$

Definition

Suppose X is a random variable with $X \in \mathbf{R}$ and probability density function

$$f(x) = \frac{1}{\sqrt{2\sigma^2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

Then X is a **normally** distributed random variable with parameters μ and σ^2 .

Equivalently, we'll write

$$X \sim \text{Normal}(\mu, \sigma^2)$$

Support for President Trump

Suppose we are interested in modeling **presidential approval**

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Proposition

Scale/Location. If $Z \sim N(0, 1)$, then $X = aZ + b$ is,

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Proof: $Z \sim N(0, 1)$ and $Y = aZ + b$, then $Y \sim N(b, a^2)$

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$$\begin{aligned}\frac{\partial F_Y(x)}{\partial x} &= \frac{\partial F_Z(\frac{x-b}{a})}{\partial x} \\ &= f_Z(\frac{x-b}{a}) \frac{1}{a} \text{ By the chain rule}\end{aligned}$$

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$$\begin{aligned}\frac{\partial F_Y(x)}{\partial x} &= \frac{\partial F_Z(\frac{x-b}{a})}{\partial x} \\ &= f_Z(\frac{x-b}{a}) \frac{1}{a} \quad \text{By the chain rule} \\ &= \frac{1}{\sqrt{2\pi}a} \exp\left[-\frac{(\frac{x-b}{a})^2}{2}\right] \quad \text{By definition of } f_Z(x) \\ &= \frac{1}{\sqrt{2\pi}a} \exp\left[-\frac{(x-b)^2}{2a^2}\right]\end{aligned}$$

Proof: $Z \sim N(0, 1)$ and $Y = aZ + b$, then $Y \sim N(b, a^2)$

So, we can work with $F_Z(\frac{x-b}{a})$.

$$\begin{aligned}\frac{\partial F_Y(x)}{\partial x} &= \frac{\partial F_Z(\frac{x-b}{a})}{\partial x} \\ &= f_Z(\frac{x-b}{a}) \frac{1}{a} \text{ By the chain rule} \\ &= \frac{1}{\sqrt{2\pi}a} \exp\left[-\frac{(\frac{x-b}{a})^2}{2}\right] \text{ By definition of } f_Z(x) \\ &= \frac{1}{\sqrt{2\pi}a} \exp\left[-\frac{(x-b)^2}{2a^2}\right] \\ &= \text{Normal}(b, a^2)\end{aligned}$$

Expectation and Variance

Assume we know:

$$E[Z] = 0$$

$$\text{Var}(Z) = 1$$

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$$= \sigma^2 + 0$$

Expectation and Variance

Assume we know:

$$\begin{aligned}E[Z] &= 0 \\ \text{Var}(Z) &= 1\end{aligned}$$

This implies that, for $Y \sim \text{Normal}(\mu, \sigma^2)$

$$\begin{aligned}E[Y] &= E[\sigma Z + \mu] \\ &= \sigma E[Z] + \mu \\ &= \mu \\ \text{Var}(Y) &= \text{Var}(\sigma Z + \mu) \\ &= \sigma^2 \text{Var}(Z) + \text{Var}(\mu) \\ &= \sigma^2 + 0 \\ &= \sigma^2\end{aligned}$$

Back To Trump

Suppose $\mu = 0.39$ and $\sigma^2 = 0.0025$

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$P(Y \geq 0.45)$

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$$P(Y \geq 0.45) = 1 - P(Y \leq 0.45)$$

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$P(Y \geq 0.45)$

$$\begin{aligned} P(Y \geq 0.45) &= 1 - P(Y \leq 0.45) \\ &= 1 - P(0.05Z + 0.39 \leq 0.45) \end{aligned}$$

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Suppose $\mu = 0.39$ and $\sigma^2 = 0.0025$

$P(Y \geq 0.45)$

$$\begin{aligned} P(Y \geq 0.45) &= 1 - P(Y \leq 0.45) \\ &= 1 - P(0.05Z + 0.39 \leq 0.45) \\ &= 1 - P\left(Z \leq \frac{0.45 - 0.39}{0.05}\right) \end{aligned}$$

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$$\begin{aligned}P(Y \geq 0.45) &= 1 - P(Y \leq 0.45) \\&= 1 - P(0.05Z + 0.39 \leq 0.45) \\&= 1 - P(Z \leq \frac{0.45 - 0.39}{0.05}) \\&= 1 - \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{6/5} \exp(-z^2/2) dz\end{aligned}$$

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The Gamma Function

Definition

Suppose $\alpha > 0$. Then define $\Gamma(\alpha)$ as

$$\Gamma(\alpha) = \int_0^{\infty} y^{\alpha-1} e^{-y} dy$$

- For $\alpha \in \{1, 2, 3, \dots\}$
 $\Gamma(\alpha) = (\alpha - 1)!$
- $\Gamma(\frac{1}{2}) = \sqrt{\pi}$

Gamma Distribution

Suppose we have $\Gamma(\alpha)$,

Gamma Distribution

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$$\frac{\Gamma(\alpha)}{\Gamma(\alpha)} = \frac{\int_0^\infty y^{\alpha-1} e^{-y} dy}{\Gamma(\alpha)}$$
$$1 = \int_0^\infty \frac{1}{\Gamma(\alpha)} y^{\alpha-1} e^{-y} dy$$

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Set $X = Y/\beta$

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$$F(x) = P(X \leq x) = P(Y/\beta \leq x)$$

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$$\begin{aligned} F(x) = P(X \leq x) &= P(Y/\beta \leq x) \\ &= P(Y \leq x\beta) \end{aligned}$$

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Set $X = Y/\beta$

$$\begin{aligned}F(x) = P(X \leq x) &= P(Y/\beta \leq x) \\ &= P(Y \leq x\beta) \\ &= F_Y(x\beta)\end{aligned}$$

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$$\begin{aligned}F(x) = P(X \leq x) &= P(Y/\beta \leq x) \\ &= P(Y \leq x\beta) \\ &= F_Y(x\beta) \\ \frac{\partial F_Y(x\beta)}{\partial x} &= f_Y(x\beta)\beta\end{aligned}$$

The result is:

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$$\frac{\Gamma(\alpha)}{\Gamma(\alpha)} = \frac{\int_0^\infty y^{\alpha-1} e^{-y} dy}{\Gamma(\alpha)}$$
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The result is:

$$f(x|\alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-x\beta}$$

Definition

Suppose X is a continuous random variable, with $X \geq 0$. Then if the pdf of X is

$$f(x|\alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-x\beta}$$

if $x \geq 0$ and 0 otherwise, we will say X is a Gamma distribution.

$$X \sim \text{Gamma}(\alpha, \beta)$$

Gamma Distribution

Suppose $X \sim \text{Gamma}(\alpha, \beta)$

Gamma Distribution

Suppose $X \sim \text{Gamma}(\alpha, \beta)$

$$E[X] = \frac{\alpha}{\beta}$$

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$$\begin{aligned} E[X] &= \frac{\alpha}{\beta} \\ \text{var}(X) &= \frac{\alpha}{\beta^2} \end{aligned}$$

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Suppose $X \sim \text{Gamma}(\alpha, \beta)$

$$\begin{aligned} E[X] &= \frac{\alpha}{\beta} \\ \text{var}(X) &= \frac{\alpha}{\beta^2} \end{aligned} \tag{0.2}$$

Suppose $\alpha = 1$ and $\beta = \lambda$. If

Gamma Distribution

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Suppose $\alpha = 1$ and $\beta = \lambda$. If

$$\begin{aligned} X &\sim \text{Gamma}(1, \lambda) \\ f(x|1, \lambda) &= \lambda e^{-x\lambda} \end{aligned}$$

We will say

Gamma Distribution

Suppose $X \sim \text{Gamma}(\alpha, \beta)$

$$\begin{aligned} E[X] &= \frac{\alpha}{\beta} \\ \text{var}(X) &= \frac{\alpha}{\beta^2} \end{aligned} \tag{0.2}$$

Suppose $\alpha = 1$ and $\beta = \lambda$. If

$$\begin{aligned} X &\sim \text{Gamma}(1, \lambda) \\ f(x|1, \lambda) &= \lambda e^{-x\lambda} \end{aligned}$$

We will say

$$X \sim \text{Exponential}(\lambda)$$

Properties of Gamma Distributions

Proposition

Suppose we have a sequence of independent random variables, with

$$X_i \sim \text{Gamma}(\alpha_i, \beta)$$

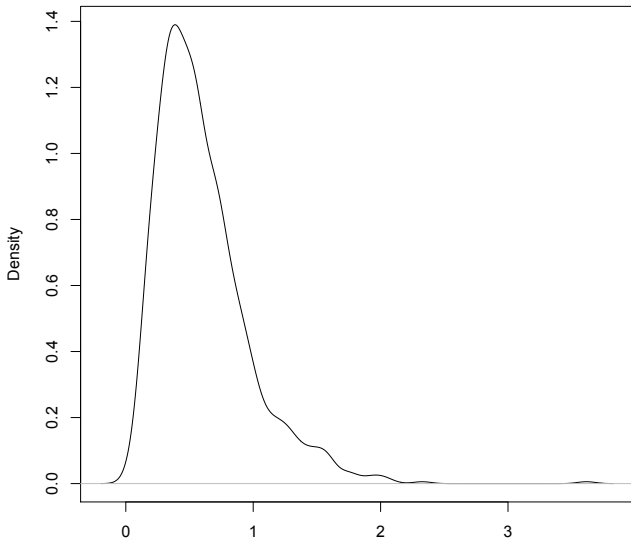
Then

$$Y = \sum_{i=1}^N X_i$$

$$Y \sim \text{Gamma}(\sum_{i=1}^N \alpha_i, \beta)$$

We can evaluate in R with `dgamma` and simulate with `rgamma`
 $X \sim \text{Gamma}(3, 5)$ and we evaluate at 3,
`dgamma(3, shape= 3, rate = 5)`
and we can simulate with
`rgamma(1000, shape = 3, rate = 5)`

`density.default(x = rgamma(1000, shape = 3, rate = 5))`



N = 1000 Bandwidth = 0.07038

χ^2 Distribution

Suppose $Z \sim \text{Normal}(0, 1)$.

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Consider $X = Z^2$

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$$F_X(x) = P(X \leq x)$$

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$$\begin{aligned} F_X(x) &= P(X \leq x) \\ &= P(Z^2 \leq x) \end{aligned}$$

χ^2 Distribution

Suppose $Z \sim \text{Normal}(0, 1)$.

Consider $X = Z^2$

$$\begin{aligned} F_X(x) &= P(X \leq x) \\ &= P(Z^2 \leq x) \\ &= P(-\sqrt{x} \leq Z \leq \sqrt{x}) \end{aligned}$$

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$$\begin{aligned} F_X(x) &= P(X \leq x) \\ &= P(Z^2 \leq x) \\ &= P(-\sqrt{x} \leq Z \leq \sqrt{x}) \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\sqrt{x}}^{\sqrt{x}} e^{-\frac{z^2}{2}} dz \end{aligned}$$

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Consider $X = Z^2$

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The pdf then is

$$\frac{\partial F_X(x)}{\partial x} = f_Z(\sqrt{x}) \frac{1}{2\sqrt{x}} + f_Z(-\sqrt{x}) \frac{1}{2\sqrt{x}}$$

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χ^2 Distribution

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χ^2 Distribution

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$$X \sim \text{Gamma}(1/2, 1/2)$$

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Then if $X = \sum_{i=1}^N Z^2$

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Then if $X = \sum_{i=1}^N Z^2$

$X \sim \text{Gamma}(n/2, 1/2)$

Definition

Suppose X is a continuous random variable with $X \geq 0$, with pdf

$$f(x) = \frac{1}{2^{n/2}\Gamma(n/2)} x^{n/2-1} e^{-x/2}$$

Then we will say X is a χ^2 distribution with n degrees of freedom. Equivalently,

$$X \sim \chi^2(n)$$

χ^2 Properties

Suppose $X \sim \chi^2(n)$

$$E[X] = E \left[\sum_{i=1}^N Z_i^2 \right]$$

$$= \sum_{i=1}^N E[Z_i^2]$$

$$\text{var}(Z_i) = E[Z_i^2] - E[Z_i]^2$$

$$1 = E[Z_i^2] - 0$$

$$E[X] = n$$

χ^2 Properties

$$\begin{aligned}\text{var}(X) &= \sum_{i=1}^N \text{var}(Z_i^2) \\ &= \sum_{i=1}^N (E[Z_i^4] - E[Z_i]^2) \\ &= \sum_{i=1}^N (3 - 1) = 2n\end{aligned}$$

We will use the χ^2 in all sorts of statistics courses

Student's t -Distribution

Definition

Suppose $Z \sim \text{Normal}(0, 1)$ and $U \sim \chi^2(n)$. Define the random variable Y as,

$$Y = \frac{Z}{\sqrt{\frac{U}{n}}}$$

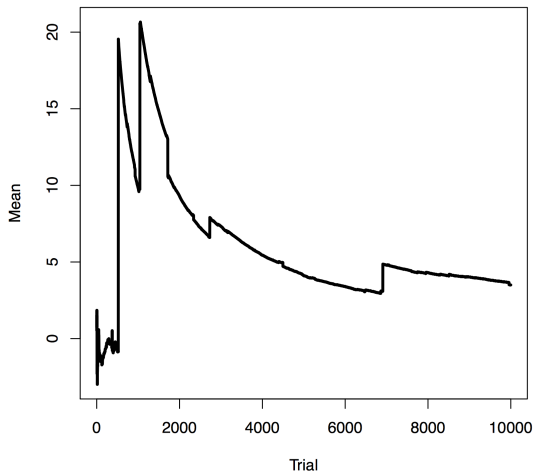
If Z and U are independent then $Y \sim t(n)$, with pdf

$$f(x) = \frac{\Gamma(\frac{n+1}{2})}{\sqrt{\pi n} \Gamma(\frac{n}{2})} \left(1 + \frac{x^2}{n}\right)^{-\frac{n+1}{2}}$$

We will use the t -distribution extensively for *test-statistics*

Student's t -Distribution, Properties

Suppose $n = 1$, **Cauchy** distribution



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$\text{var}(X) = \text{undefined}$

If $X \sim t(2)$

$E[X] = 0$

$\text{var}(X) = \text{undefined}$

Student's t -Distribution, Properties

Suppose $n > 2$, then

$$\text{var}(X) = \frac{n}{n-2}$$

As $n \rightarrow \infty$ $\text{var}(X) \rightarrow 1$.

Using the t-Distribution

We often construct **test-statistics**

Suppose we take N iid draws,

$$X_i \sim \text{Normal}(\mu, \sigma^2)$$

Define our data set $\mathbf{x} = (x_1, \dots, x_N)$

Calculate:

$$\bar{x} = \sum_{i=1}^N \frac{x_i}{N}$$

$$s^2 = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2$$

Estimators

- Our data are realizations from a sample space.

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Estimators

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$\bar{x}(X)$ is a function called the sample mean, and it produces estimates of the true mean (μ)

$s^2(X)$ is a function called the sample variance, and it produces estimates of the true variance (σ^2).

Sometimes we will use little hats, to indicate that it is a function of a particular realization of the data.

Estimators

- Because estimators are random variables, they have a distribution.

Estimators

- Because estimators are random variables, they have a distribution.
- Our knowledge about the underlying world that produced our data is governed by these distributions.

Using the T-Distribution

Define

$$t = \frac{\bar{x} - \mu}{s/\sqrt{N}}$$
$$t \sim \text{Student's } t(N - 1)$$